

1 Detector

Person boxes
PPE evidence
Confidence scores

2 Assignment

Geometric support
One box, one worker
No shared evidence

3 Worker state

Helmet and vest
Safe / unsafe / review
Threshold t

4 Calibration

Ordered exact tests
Stop at first failure
Maximize coverage

Calibration split

Filename-group-disjoint roles
Fixed grid T
Model-conditional UCB

SAFE

UNSAFE

REVIEW

Review-all fallback

No retained threshold
All cases routed
to review

